

Emin Erkan YILDIRIM

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Education

Dokuz Eylul University – MS in Electrical and Electronics Engineering	February 2025 – Present
Dokuz Eylul University – BS in Electrical and Electronics Engineering (GPA: 3.0/4.0)	September 2019 – June 2024

Experience

Laboratory Specialist R&D , Vestel Beyaz Eşya – Manisa, Türkiye	October 2024 – Present
- Verified electronic control boards used in tumble dryers to ensure optimal functionality, performance, and compliance with IEC 60335 standards. - Prepared engineering requirements and technical specification documents. - Conducted benchmark studies and analyzed competitor products to identify new components suitable for integration. - Made design decisions to ensure product safety and regulatory compliance. - Communicated with new component suppliers to achieve advantages in cost and quality.	
Embedded Software Engineering Intern , Baykar Technologies – Istanbul, Türkiye	July 2023 – August 2023
Prepared and delivered a technical presentation on a communication protocol, including detailed analysis and comparison with alternative communication protocols.	
Research and Development Intern , Guleryuz Karoser Otomotiv A.Ş. – Bursa, Türkiye	July 2022 – August 2022
- Inspected CAN bus scheme placement in an electrical system. - Standardized signal naming on the model to accelerate production line placement. - Determined required cable lengths for CAN bus lines using the system model.	

Projects

Track Shirt for Tracking Athlete Performance (TUBITAK 1001)

- Designed a PCB in KiCad for textile integration using conductive yarn.
- Developed RSL10 application firmware in C for sensor transmission using BLE GATT notifications.
- Implemented acquisition and wireless transmission of acceleration and ECG data.

User Interface and Data Logger for Solar Car

- Developed a C# interface to read UART telemetry from a USB-connected LoRa receiver.
- Implemented real-time graph visualization using the LiveCharts library.
- Validated packet checksums and logged telemetry to CSV during an actual race for over one hour, without observed crashes or freezes.

Four-legged Walking Robot Simulation

- Designed and programmed an autonomous quadruped robot, implementing locomotion algorithms in Python using Webots.

Extracurricular Activities

Solar Team Solaris - Electronic Subgroup Member	October 2021 – October 2022
- Reverse-engineered incoming LoRa data using HTerm to analyze communication frames. - Designed and developed a C# interface to retrieve and display real-time data from a solar-powered racing car based on the analysis. - Developed CAN communication software in C on an STM32 microcontroller and built circuitry for communication testing and validation.	

Skills

Programming: C, C#, Python, SystemVerilog, MATLAB
Communication Protocols: TCP/IP, UDP, SPI, UART, CAN, I2C, MIL-STD-1553
Software: LTspice, KiCAD, QUCS, Webots
Languages: English (Advanced), Turkish (Native)